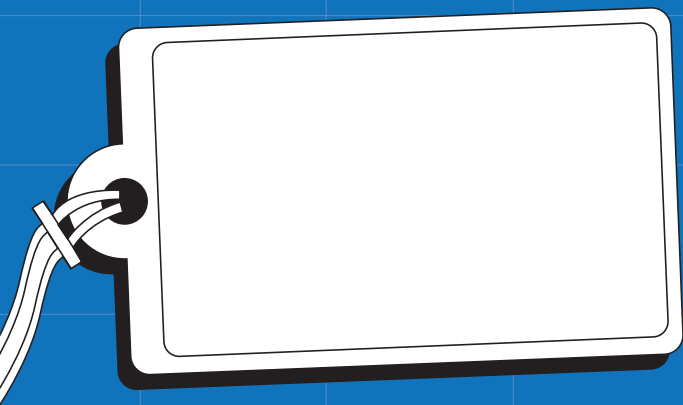




SURDS
&
INDICES

SURDS & INDICES

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SURDS AND INDICES

Sometimes just writing variables or 'pronumerals' in mathematical expressions is not enough. Sometimes the value of the variable is needed. An equation is used to find the value of a variable.



Answer these questions, *before* working through the chapter.

I used to think:

What is a rational number?

Which one of these is true: $\sqrt{4} \times \sqrt{100} = \sqrt{400}$ or $\sqrt{4} + \sqrt{100} = \sqrt{104}$?

Why are $\frac{1}{\sqrt{2}}$ and $\frac{\sqrt{2}}{2}$ equal?

Answer these questions, *after* working through the chapter.

But now I think:

What is a rational number?

Which one of these is true: $\sqrt{4} \times \sqrt{100} = \sqrt{400}$ or $\sqrt{4} + \sqrt{100} = \sqrt{104}$?

Why are $\frac{1}{\sqrt{2}}$ and $\frac{\sqrt{2}}{2}$ equal?



What do I know now that I didn't know before?

All real numbers can be classified as either *rational* or *irrational*.

What is a Rational Number

Rational numbers are any numbers that can be written as a fraction $\frac{a}{b}$ where a and b are integers and $b \neq 0$. This means that rational numbers include:

- 1 Obviously all fractions since by definition they are written as $\frac{a}{b}$ where a and b are integers and $b \neq 0$.
- 2 All integers since any integer can be written as a fraction with denominator 1.
- 3 Any decimal number that terminates (eg. $0.5 = \frac{1}{2}$) or recur eg. ($0.\dot{7} = \frac{7}{9}$)

If a number can be written as a terminating or recurring decimal, then it is a rational number. If the number does not terminate or repeat, then it is not rational. These numbers are called **irrational** numbers.

Are these numbers rational or irrational?

a 3

This can be written as $\frac{3}{1}$
 $\therefore 3$ is a **rational** number

b $0.\dot{4}\dot{5}$

This is a recurring decimal
 $\therefore 0.\dot{4}\dot{5}$ is a **rational** number
 $0.\dot{4}\dot{5}$ can be written as $\frac{45}{99}$

c π

$\pi = 3.141592654\dots$
 This is a nonterminating nonrepeating decimal
 $\therefore \pi$ is an **irrational** number

d 0.3256

This decimal terminates
 $\therefore 0.3256$ is a **rational** number
 0.3256 can be written as $\frac{3256}{10000}$

Surds

The symbol $\sqrt{x}$ means the positive square root of x . There are three cases for a number under $\sqrt{\quad}$.

- If x is a perfect square then the square root will be a rational number. (eg. $\sqrt{4} = 2$)
- If x is not a perfect square, then the square root is irrational. For example $\sqrt{7}$ is irrational. These types of irrational numbers are called surds. It's easy to see $\sqrt{7}$ is irrational, since $\sqrt{7} = 2.6457513\dots$ is a non-recurring and non-terminating decimal.
- If x is negative, then $\sqrt{x}$ is **undefined** because there is no square root of a negative number which is a real number.

Why are ALL integers rational numbers? Give an example.

Identify whether these numbers are rational or irrational. Give a reason.

a $\frac{5}{11}$

b 10

c 5.3472294401...

d 0.3333333...

e 0.324324324324...

f 0.43554

Identify whether each of the following numbers is rational, irrational or is undefined (no real solution). Give a reason

a $\sqrt{-10}$

b $\sqrt{12}$

c $\sqrt{9}$

d $\sqrt{100}$

e $\sqrt{5}$

f $\sqrt{36}$

g $\sqrt{0}$

h $\sqrt{-100}$

Between which two integers are these surds?

a $\sqrt{11}$

b $\sqrt{40}$

c $\sqrt{111}$

d $\sqrt{93}$

Multiplying Surds

Look at this example:

$$\sqrt{4} \times \sqrt{16} = 2 \times 4 = 8 = \sqrt{64} = \sqrt{(4 \times 16)}$$

So it's easy to see $\sqrt{4} \times \sqrt{16} = \sqrt{(4 \times 16)}$. This is true for all numbers and the rule for multiplying surds is

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

and

$$\sqrt{a} \times \sqrt{a} = (\sqrt{a})^2 = a$$

Here are some examples:

a $\sqrt{5} \times \sqrt{20}$	b $\sqrt{6} \times \sqrt{8} \times \sqrt{3}$	c $\sqrt{6} \times \sqrt{6}$	d $\sqrt{7} \times \sqrt{6}$
$= \sqrt{5 \times 20}$	$= \sqrt{6 \times 8 \times 3}$	$= \sqrt{6 \times 6}$	$= \sqrt{7 \times 6}$
$= \sqrt{100}$	$= \sqrt{144}$	$= \sqrt{36}$	$= \sqrt{42}$
$= 10$	$= 12$	$= 6$	(Can't be simplified further)

The multiplication rule can also be used to simplify surds. If the number under the $\sqrt{\quad}$ can be written as a product of a perfect square and a non perfect square then the surd can be simplified. Here is an example:

$$\sqrt{300} = \sqrt{100 \times 3} = \sqrt{100} \times \sqrt{3} = 10\sqrt{3}$$

Perfect square $\times$ non perfect square
Simplest surd form

Look for multiples of 4, 9, 16, 25, 36 etc, since these are perfect squares.

Simplify these surds:

a $\sqrt{8}$	b $\sqrt{75}$	c $\sqrt{147}$
$= \sqrt{4 \times 2}$	$= \sqrt{25 \times 3}$	$= \sqrt{49 \times 3}$
$= \sqrt{4} \times \sqrt{2}$	$= \sqrt{25} \times \sqrt{3}$	$= \sqrt{49} \times \sqrt{3}$
$= 10\sqrt{2}$	$= 5\sqrt{3}$	$= 7\sqrt{3}$

Dividing Surds

Look at this example:

$$\sqrt{\frac{16}{4}} = \sqrt{4} = 2 = \frac{4}{2} = \frac{\sqrt{16}}{\sqrt{4}}$$

So it's easy to see that $\sqrt{\frac{16}{4}} = \frac{\sqrt{16}}{\sqrt{4}}$. This is true for all numbers and the rule for dividing surds is

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

Here are some examples:

a $\sqrt{80} \div \sqrt{20}$

$$= \frac{\sqrt{80}}{\sqrt{20}} = \sqrt{\frac{80}{20}}$$

$$= \sqrt{4}$$

$$= 2$$

b $\sqrt{\frac{16}{9}}$

$$= \frac{\sqrt{16}}{\sqrt{9}}$$

$$= \frac{4}{3}$$

c $\sqrt{\frac{11}{25}}$

$$= \frac{\sqrt{11}}{\sqrt{25}}$$

$$= \frac{\sqrt{11}}{5}$$

What if There is a Number Outside the Surd?

If there are numbers outside of the surd then they are multiplied or divided separately.

Here are some examples:

a $3\sqrt{6} \times 2\sqrt{24}$

$$= 3 \times 2 \times \sqrt{6 \times 24}$$

$$= 6\sqrt{144}$$

$$= 6(12)$$

$$= 72$$

b $8\sqrt{12} \times 12\sqrt{8}$

$$= 8 \times 12 \times \sqrt{12 \times 8}$$

$$= 96\sqrt{96}$$

$$= 96\sqrt{16 \times 6}$$

$$= 96(\sqrt{16} \times \sqrt{6})$$

$$= 96(4\sqrt{6}) = 384\sqrt{6}$$

c $6\sqrt{8} \div 3\sqrt{4}$

$$= \frac{6\sqrt{8}}{3\sqrt{4}}$$

$$= 2\frac{\sqrt{8}}{\sqrt{4}}$$

$$= 2\sqrt{\frac{8}{4}}$$

$$= 2\sqrt{2}$$

Adding and Subtracting Surds

Look at these examples:.

$$\begin{array}{lcl} \sqrt{9} + \sqrt{4} & & \sqrt{9+4} \\ = 3 + 2 & \text{and} & = \sqrt{13} \\ = 5 & & = 3.6055... \end{array}$$

So its easy to see that $\sqrt{9} + \sqrt{4}$ is not equal to $\sqrt{9+4}$. This is true for all numbers, so for any numbers a and b

$$\sqrt{a} + \sqrt{b} \neq \sqrt{a+b}$$

↑
This sign means "not equal to"

The same can be said for subtraction

$$\sqrt{a} - \sqrt{b} \neq \sqrt{a-b}$$

Like Terms

Surds can be added and subtracted if they are like terms. Like terms have the same number under the $\sqrt{\quad}$.

Here are some examples:

Simplify the following expressions:

a $\sqrt{3} + 2\sqrt{3}$

↑ ↑
Like Terms

$= 3\sqrt{3}$

b $4\sqrt{2} + 3\sqrt{5} - \sqrt{2} + \sqrt{5} - 5\sqrt{5}$

↑ ↑ ↑ ↑
Like Terms Like Terms

$= 3\sqrt{2} - \sqrt{5}$

↑ ↑
These are unlike terms and cannot be simplified further

Sometimes the surds will have to be converted to simplest surd form before like terms can be found.

Convert the following to simplest surd form and then simplify:

a $\sqrt{63} - \sqrt{7} + \sqrt{28}$

Convert to simplest surd form

$$\begin{aligned} &= \sqrt{9 \times 7} - \sqrt{7} + \sqrt{4 \times 7} \\ &= 3\sqrt{7} - \sqrt{7} + 2\sqrt{7} \end{aligned}$$

Simplify like terms

$$= 4\sqrt{7}$$

b $\sqrt{72} + \sqrt{27} - \sqrt{75} + \sqrt{8}$

Convert to simplest surd form

$$\begin{aligned} &= \sqrt{36 \times 2} + \sqrt{9 \times 3} - \sqrt{25 \times 3} + \sqrt{4 \times 2} \\ &= 6\sqrt{2} + 3\sqrt{3} - 5\sqrt{3} + 2\sqrt{2} \end{aligned}$$

Simplify like terms

$$= 8\sqrt{2} - 2\sqrt{3}$$

Find the following products. Simplify to an integer if possible:

a $4x + 3 = 23$

b $(\sqrt{11})^2$

c $\sqrt{18} \times \sqrt{8}$

d $\sqrt{5} \times \sqrt{6}$

e $\sqrt{5} \times \sqrt{10} \times \sqrt{2}$

f $\sqrt{3} \times \sqrt{6} \times \sqrt{5}$

Find the following quotients. Simplify to an integer if possible:

a $\sqrt{\frac{144}{9}}$

b $\sqrt{\frac{121}{9}}$

c $\sqrt{\frac{100}{20}}$

d $\sqrt{\frac{3}{64}}$

e $\sqrt{\frac{100}{49}}$

f $\sqrt{\frac{8}{81}}$

Write the following in simplest surd form:

a $\sqrt{18}$

b $\sqrt{24}$

c $\sqrt{80}$

d $\sqrt{150}$

e $\sqrt{1200}$

f $\sqrt{147}$

Simplify the following into simplest surd form:

a $\sqrt{10} \times \sqrt{8}$

b $\sqrt{5} \times \sqrt{40}$

c $2\sqrt{2} \times \sqrt{75}$

d $4\sqrt{10} \times 2\sqrt{5}$

e $3\sqrt{\frac{9}{4}}$

f $6\sqrt{25} \div 3\sqrt{12}$

Find the like terms in each of the following:

a $3\sqrt{3}, \sqrt{3}, \sqrt{2}$

b $\sqrt{7}, 2\sqrt{11}, -\sqrt{11}, 4\sqrt{7}$

c $2\sqrt{5}, 5\sqrt{2}, 7\sqrt{2}, -2\sqrt{2}, \sqrt{5}$

d $\sqrt{a}, a\sqrt{2}, 3\sqrt{2}, a\sqrt{a}$

Simplify these expressions, if possible:

a $7\sqrt{3} - 5\sqrt{3}$

b $2\sqrt{5} + 4\sqrt{5} - 7\sqrt{5}$

c $6\sqrt{2} + 5\sqrt{3} - 4\sqrt{2}, -7\sqrt{3}$

d $3\sqrt{5} + 5\sqrt{3} - 2\sqrt{5} + 4\sqrt{5}$

Find the simplest surd form and then simplify the following as much as possible:

a $\sqrt{18} - \sqrt{2} + \sqrt{50}$

b $\sqrt{100} + \sqrt{75} - \sqrt{48}$

c $\sqrt{112} - \sqrt{6} - \sqrt{7} + \sqrt{150}$

d $\sqrt{40} + \sqrt{1000} - \sqrt{20} - \sqrt{10}$

Brackets with Surds

The 'distributive law' says that if a term multiplies a bracket, then it must multiply all the terms in the bracket.

For example:

$$a(x + y) = ax + ay$$

The same is true if any of the terms contain surds. Here are some examples:

Simplify the following:

a $\sqrt{3}(\sqrt{5} + \sqrt{7})$

$$= \sqrt{3} \times \sqrt{5} + \sqrt{3} \times \sqrt{7}$$

$$= \sqrt{15} + \sqrt{21}$$

b $4\sqrt{5}(2\sqrt{3} + \sqrt{7})$

$$= 4\sqrt{5} \times 2\sqrt{3} + 4\sqrt{5} \times \sqrt{7}$$

$$= 8\sqrt{15} + 4\sqrt{35}$$

Binomial Products with Surds

A 'binomial product' occurs when brackets with two terms are multiplied together. For example

$$(a + b)(x + y) = ax + ay + bx + by$$

The same is true if any of the terms contain surds. Here are some examples

Simplify the following:

a $(\sqrt{2} + \sqrt{5})(\sqrt{11} - \sqrt{7})$

$$= \sqrt{22} - \sqrt{14} + \sqrt{55} - \sqrt{35}$$

b $(2\sqrt{6} + \sqrt{15})(\sqrt{6} + 3\sqrt{5})$

$$= 12 + 6\sqrt{30} + \sqrt{90} + 3\sqrt{75}$$

$$= 12 + 6\sqrt{30} + 3\sqrt{10} + 15\sqrt{3}$$

(simplest surd form)

If a bracket is squared, simply multiply the bracket with itself. For example

$$(2\sqrt{2} + \sqrt{5})^2 = (2\sqrt{2} + \sqrt{5})(2\sqrt{2} + \sqrt{5})$$

and now the the method in the previous examples can be used.

Conjugates

The binomials $\sqrt{3} + \sqrt{2}$ and $\sqrt{3} - \sqrt{2}$ differ only in the sign of the second term. These binomials are called **conjugates** of each other. When finding conjugates, remember that **only the sign of the second term changes, not the signs of both terms.**

Find the Conjugate of the following:

a $\sqrt{2} + \sqrt{5}$

The conjugate is $\sqrt{2} - \sqrt{5}$

b $3 - \sqrt{7}$

The conjugate is $3 + \sqrt{7}$

c $3\sqrt{2} + 4\sqrt{3}$

The conjugate is $3\sqrt{2} - 4\sqrt{3}$

A Shortcut to Multiply a Binomial with its Conjugate

The product of $(a + b)$ and its conjugate $(a - b)$ can be simplified like this:

$$\begin{aligned}(a + b)(a - b) \\&= a^2 - ab + ab - b^2 \\&= a^2 - b^2\end{aligned}$$

This is just the difference of the squares of the terms in the original bracket. Here are some examples with surds

Find the following products:

a $(\sqrt{5} + 3\sqrt{2})(\sqrt{5} - 3\sqrt{2})$

$$= (\sqrt{5})^2 - (3\sqrt{2})^2$$

$$= 5 - 18$$

$$= -13$$

b $(5\sqrt{3} - 2\sqrt{7})(5\sqrt{3} + 2\sqrt{7})$

$$= (5\sqrt{3})^2 - (2\sqrt{7})^2$$

$$= 75 - 28$$

$$= 47$$

The product of a surd and its conjugate will always be the difference of the square of the first term and the square of the second term

$$(a + b)(a - b) = (a^2 - b^2)$$

Since the square of a surd is equal to the number in the surd $(\sqrt{a})^2 = a$, the product of a surd and its conjugate is **always a rational number.**

Simplify these products to simplest surd form:

a $\sqrt{5}(2\sqrt{3} + \sqrt{7})$

b $\sqrt{3}(2\sqrt{2} - 7\sqrt{5})$

c $\sqrt{11}(3\sqrt{11} + 4\sqrt{2})$

d $3\sqrt{10}(6\sqrt{2} - 8\sqrt{7})$

e $3\sqrt{7}(7\sqrt{3} + 2\sqrt{15})$

f $2\sqrt{2}(3\sqrt{32} + \sqrt{72} - 2\sqrt{7})$

g $\sqrt{x}(3\sqrt{2} + 2\sqrt{5})$

h $\sqrt{p}(2\sqrt{p} - 4\sqrt{10})$

i $2\sqrt{a}(5\sqrt{6} - \sqrt{a})$

j $3\sqrt{m}(m\sqrt{3} + 2\sqrt{m})$

Find these binomial products and simplify to simplest surd form:

a $(\sqrt{2} + \sqrt{3})(\sqrt{7} + 6)$

b $(4\sqrt{3} - \sqrt{5})(\sqrt{32} + \sqrt{20})$

c $(3\sqrt{7} - 2\sqrt{10})(\sqrt{5} + \sqrt{11})$

d $(\sqrt{7} + 5\sqrt{2})(6\sqrt{5} - \sqrt{3})$

e $(8\sqrt{5} - 3\sqrt{2})(7\sqrt{10} - 4\sqrt{2})$

f $(3\sqrt{6} + 7\sqrt{5})(2\sqrt{11} - 6\sqrt{5})$

g $(\sqrt{5} + \sqrt{7})^2$

h $(2\sqrt{3} - 11\sqrt{5})^2$

i $(2\sqrt{3} + 5\sqrt{11})(2\sqrt{3} - 5\sqrt{11})$

j $(4\sqrt{3x} + 3\sqrt{3})(4\sqrt{x} - 3\sqrt{3})$

Write down the conjugate of the following:

a $2\sqrt{5} + \sqrt{6}$

b $\sqrt{10} - 3\sqrt{7}$

c $5\sqrt{11} - 2\sqrt{2}$

d $6\sqrt{7} + 8\sqrt{10}$

Find the product of the following with its conjugate

a $4 - \sqrt{5}$

b $\sqrt{3} + \sqrt{6}$

c $5\sqrt{5} + \sqrt{7}$

d $6\sqrt{2} - 3\sqrt{10}$

e $2\sqrt{11} + 4\sqrt{15}$

f $\sqrt{a} + \sqrt{b}$

g $2\sqrt{m} - 3\sqrt{n}$

h $a\sqrt{b} + c\sqrt{d}$

Rationalising the Denominator

- Fractions with **rational** denominators (like $\frac{\sqrt{2}}{2}$) are easier to use in calculations than fractions with irrational denominators (like $\frac{1}{\sqrt{2}}$). It's important to know how to change from an irrational denominator to a rational denominator.
- To change a fraction with an **irrational denominator** to an *equal* fraction with a **rational denominator**, multiply the top and bottom of the fraction by the surd in the denominator.

Change these fractions into EQUAL fractions with a rational denominator

a $\frac{1}{\sqrt{2}}$ ← Irrational Denominator

The surd in the denominator is $\sqrt{2}$

∴ multiply top and bottom by $\sqrt{2}$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{2}}{2} \leftarrow \text{Irrational Denominator}$$

b $\frac{\sqrt{3}}{2\sqrt{5}}$

The surd in the denominator is $\sqrt{5}$

∴ multiply top and bottom by $\sqrt{5}$ (not $2\sqrt{5}$)

$$= \frac{\sqrt{3}}{2\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$$

$$= \frac{\sqrt{15}}{10}$$

Multiplying the top and bottom by the same number is the same as multiplying the fraction by 1. So the original fraction and the 'rationalised fraction' are equal.

Since the fractions with irrational denominators have been changed to equal fractions with a rational denominator, this method is called '**rationalising the denominator.**'

Rationalise each denominator and then simplify

Step 1: Rationalise the denominator of each fraction

$$\frac{2}{\sqrt{5}} + \frac{7}{\sqrt{10}}$$

$$= \frac{2}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} + \frac{7}{\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}}$$

$$= \frac{2\sqrt{5}}{5} + \frac{7\sqrt{10}}{10}$$

Step 2: Find a common denominator and simplify

$$= \frac{4\sqrt{5} + 7\sqrt{10}}{10}$$

What if the Denominator Contains an Irrational Binomial?

- Sometimes the denominator has a binomial containing surds, like $\frac{3}{\sqrt{2} + \sqrt{5}}$.
- To rationalise the denominator, multiply the top and bottom of the fraction by the denominator's conjugate.

Rationalise the denominator of the following:

a $\frac{1}{\sqrt{2}}$

The denominator conjugate is $7 + \sqrt{2}$

∴ multiply top and bottom by $7 + \sqrt{2}$

$$\begin{aligned} &= \frac{4}{7 - \sqrt{2}} \times \frac{7 + \sqrt{2}}{7 + \sqrt{2}} \\ &= \frac{28 + 4\sqrt{2}}{47} \end{aligned}$$

b $\frac{2\sqrt{3}}{\sqrt{10} + 2\sqrt{2}}$

The denominator's conjugate is $\sqrt{10} - 2\sqrt{2}$

∴ multiply top and bottom by $\sqrt{10} - 2\sqrt{2}$

$$\begin{aligned} &= \frac{\sqrt{3}}{2\sqrt{5}} \times \frac{\sqrt{10} - 2\sqrt{2}}{\sqrt{10} - 2\sqrt{2}} \\ &= \frac{2\sqrt{30} - 4\sqrt{6}}{2} \end{aligned}$$

Rationalise each denominator and then simplify

Step 1: Rationalise the denominator of each fraction

$$\begin{aligned} &\frac{2\sqrt{2}}{\sqrt{7} - \sqrt{5}} + \frac{3}{\sqrt{5} - 1} \\ &= \frac{2\sqrt{2}}{\sqrt{7} - \sqrt{5}} \times \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} + \sqrt{5}} + \frac{3}{\sqrt{5} - 1} \times \frac{\sqrt{5} + 1}{\sqrt{5} + 1} \\ &= \frac{2\sqrt{2}(\sqrt{7} + \sqrt{5})}{(\sqrt{7})^2 - (\sqrt{5})^2} + \frac{3(\sqrt{5} + 1)}{(\sqrt{5})^2 - (1)^2} \\ &= \frac{2\sqrt{14} + 2\sqrt{10}}{2} + \frac{3\sqrt{5} + 3}{4} \end{aligned}$$

Step 2: Find a common denominator and simplify

$$\begin{aligned} &= \frac{2(2\sqrt{14} + 2\sqrt{10}) + 3\sqrt{5} + 3}{4} \\ &= \frac{(4\sqrt{14} + 4\sqrt{10}) + 3\sqrt{5} + 3}{4} \end{aligned}$$

Surds can be written as Fractional Indices

'indices' is another word for 'exponents' or 'powers'

- According to the laws of indices $(x^{\frac{1}{2}})^2 = x^{\frac{1}{2} \times 2} = x$
- According to the laws of surds $(\sqrt{x})^2 = x$
- This means that $x^{\frac{1}{2}} = \sqrt{x}$

If the index of any number is $\frac{1}{2}$ then the square root is found.

Simplify:

a $4^{\frac{1}{2}}$

$$= \sqrt{4}$$

$$= 2$$

b $64^{\frac{1}{2}}$

$$= \sqrt{64}$$

$$= 8$$

- According to the laws of indices $(x^{\frac{1}{3}})^3 = x^{\frac{1}{3} \times 3} = x$
- According to the laws of surds $(\sqrt[3]{x})^3 = x$
- This means that $x^{\frac{1}{3}} = \sqrt[3]{x}$

If the index of any number is $\frac{1}{3}$ then the cube root is found.

Simplify:

a $8^{\frac{1}{3}}$

$$= \sqrt[3]{8} = \sqrt[3]{2^3}$$

$$= 2$$

b $1000^{\frac{1}{3}}$

$$= \sqrt[3]{1000} = \sqrt[3]{10^3}$$

$$= 10$$

So, from the above it is easy to see:

- $x^{\frac{1}{2}}$ means the **square root** of x .
- $x^{\frac{1}{3}}$ means the **cube root** of x .

What about if the index is $\frac{1}{4}$ or $\frac{1}{5}$?

- In a similar way, a rule could be found for $x^{\frac{1}{4}}$, $x^{\frac{1}{5}}$, etc. The rule for these indices is

$$x^{\frac{1}{n}} = \sqrt[n]{x}$$

- In addition

$$x^{\frac{m}{n}} = (x^{\frac{1}{n}})^m = (\sqrt[n]{x})^m \quad \text{and} \quad x^{\frac{m}{n}} = (x^m)^{\frac{1}{n}} = (\sqrt[n]{x^m})$$

So the rule for any number which has any fraction $\frac{m}{n}$ as the index is

$$x^{\frac{m}{n}} = (\sqrt[n]{x})^m = \sqrt[n]{x^m}$$

Here are some examples

Simplify:

a $6^{\frac{1}{2}}$
 $= \sqrt{6}$

b $a^{\frac{3}{4}}$
 $= \sqrt[4]{a^3} \text{ or } (\sqrt[4]{a})^3$

c $7^{-\frac{1}{2}}$
 $= \frac{1}{7^{\frac{1}{2}}}$
 $= \frac{1}{\sqrt{7}}$

remember: $x^{-n} = \frac{1}{x^n}$

Write in index form:

a $\sqrt{11}$
 $= 11^{\frac{1}{2}}$

b $\sqrt[3]{a^5}$
 $= a^{\frac{5}{3}}$

c $(\sqrt[4]{5})^2$
 $= \sqrt[4]{5^2} = \sqrt[4]{25}$
 $= 25^{\frac{1}{4}}$

Evaluate without using a calculator:

a $4^{\frac{3}{2}}$
 $= \sqrt{4^3}$
 $= \sqrt{64}$
 $= 8$

b $4\sqrt{5^8}$
 $= 5^{\frac{8}{4}}$
 $= 5^2$
 $= 25$

c $\sqrt[6]{16^3}$
 $= 16^{\frac{3}{6}}$
 $= 16^{\frac{1}{2}} = \sqrt{16}$
 $= 4$

Rationalise the denominators of these fractions, and simplify as much as possible

a $\frac{6}{\sqrt{12}}$

b $\frac{\sqrt{3}}{\sqrt{5}}$

c $\frac{3\sqrt{6}}{\sqrt{7}}$

d $\frac{5}{2\sqrt{10}}$

e $\frac{\sqrt{3} + 1}{\sqrt{12}}$

f $\frac{3\sqrt{12}}{\sqrt{6}}$

g $\frac{4\sqrt{12} + \sqrt{2}}{\sqrt{2}}$

h $\frac{5\sqrt{7} + 3\sqrt{2}}{8\sqrt{5}}$

i $\frac{5}{\sqrt{m}}$

j $\frac{2\sqrt{3x}}{7\sqrt{x}}$

Simplify these as much as possible, by rationalising the denominator of each fraction

a $\frac{2}{\sqrt{3}} + \frac{3}{\sqrt{2}}$

b $\frac{3}{\sqrt{7}} + \frac{\sqrt{6}}{\sqrt{5}}$

c $\frac{\sqrt{12}}{\sqrt{3}} - \frac{\sqrt{2}}{\sqrt{8}}$

d $\frac{5\sqrt{20}}{\sqrt{10}} + \frac{\sqrt{3}}{\sqrt{5}}$

Rationalise the denominator of the following:

a $\frac{2}{\sqrt{2} + \sqrt{7}}$

b $\frac{4\sqrt{6}}{2\sqrt{3} - \sqrt{5}}$

c $\frac{2 - \sqrt{5}}{3\sqrt{7} + 2\sqrt{10}}$

d $\frac{2\sqrt{3} + 3\sqrt{12}}{\sqrt{5} - 3\sqrt{2}}$

Simplify the following as much as possible, by rationalising the denominator of each fraction

a $\frac{3}{\sqrt{5} + 2\sqrt{2}} + \frac{2}{\sqrt{7} + \sqrt{3}}$

b $\frac{\sqrt{2}}{\sqrt{3} - \sqrt{10}} - \frac{1}{3\sqrt{5} - \sqrt{2}}$

Write the following in surd form

a $5^{\frac{1}{4}}$

b $3^{\frac{2}{3}}$

c $x^{\frac{4}{3}}$

d $(5x)^{\frac{1}{2}}$

e $100^{\frac{5}{7}}$

f $(4q)^{\frac{7}{4}}$

Write the following index form

a $\sqrt{7}$

b $\sqrt{7^3}$

c $\sqrt[3]{5}$

d $\sqrt[6]{3^8}$

e $\sqrt{y^5}$

f $\sqrt[3]{p^4}$

g $\sqrt[n]{4}$

h $\sqrt[p]{x^q}$

Find the value of the following (without a calculator)

a $4^{\frac{1}{2}}$

b $27^{\frac{1}{3}}$

c $16^{\frac{3}{2}}$

d $9^{\frac{3}{2}}$

e $125^{\frac{2}{3}}$

f $100^{\frac{5}{2}}$

g $10000^{\frac{3}{4}}$

h $64^{\frac{1}{6}} \times 32^{\frac{1}{5}}$

i $\left(\frac{8}{343}\right)^{\frac{1}{3}}$

j $(27^{\frac{2}{3}} \times 32^{\frac{2}{3}})^{\frac{1}{2}}$

Show that $\frac{2}{\sqrt{7}+4} - \frac{2}{\sqrt{7}-4}$ is a rational number.

